

RISK AND VULNERABILITY METRICS

From the Conditional Density to a Multidimensional Risk Profile

Methodological section continuing the distributional fitting framework of the previous chapter. Having recovered a continuous conditional Johnson SU density at every forecast origin, we now extract from it the risk and vulnerability metrics that constitute the diagnostic output of the Oil-at-Risk framework.

1 Motivation: A Multidimensional View of Oil Price Risk

The distribution fitting step described in the previous chapter delivers, for every forecast origin t and horizon h , a four-parameter conditional density $f_{t+h}(x \mid \hat{\Theta}_t)$ belonging to the Johnson SU family. This density encodes everything the model knows about the future distribution of Brent returns given the current state of the world: the level of geopolitical risk, the recent volatility regime, the strength of the dollar, and the state of global trade. But a density, by itself, is not a decision tool. A risk manager facing a potential supply disruption in the Strait of Hormuz does not consult a probability density function; she asks how large her worst-case loss might be, how severe the tail would be if that worst case materialises, and whether the current distribution has shifted in a way that signals an abnormal accumulation of downside risk relative to historical norms.

These are fundamentally different questions, and they require different metrics. The first is answered by Value-at-Risk, a quantile of the conditional distribution. The second is answered by Expected Shortfall, the conditional expectation in the tail beyond that quantile. The third is answered by a measure of distributional distance, specifically the Kullback–Leibler divergence between the conditional forecast density and a reference density that represents the market’s normal state. Together with distributional skewness, which tracks the directional asymmetry of the forecast distribution over time, these four metrics provide a multidimensional risk profile that no single statistic can capture.

The motivation for computing all four metrics, rather than relying on any single one, is grounded in the economic structure of crude oil markets. Oil prices are subject to at least three distinct types of risk that manifest differently in the conditional distribution.

A supply disruption driven by geopolitical escalation fattens the right tail of the return distribution as markets price in the possibility of a sharp price spike, while leaving the left tail relatively stable. A demand collapse driven by a global recession fattens the left tail as markets anticipate a sustained price decline, while the right tail compresses. A regime of elevated uncertainty, such as the months surrounding the onset of the COVID-19 pandemic in early 2020, can fatten both tails simultaneously as the market’s inability to resolve the direction of the next large move widens the entire distribution. Value-at-Risk captures the threshold of these tails; Expected Shortfall captures their depth; Kullback–Leibler divergence captures their departure from normality; skewness captures their asymmetry. The Oil-at-Risk framework reports all four, for both tails, at every forecast origin in the out-of-sample evaluation.

The remainder of this chapter proceeds as follows. Section 2 introduces Value-at-Risk and its computation from both the conditional density and from historical simulation, including the time-aggregation conventions mandated by Basel III. Section 3 extends the analysis to Expected Shortfall, the coherent risk measure that addresses the well-known limitations of Value-at-Risk in the tails. Section 4 develops the Kullback–Leibler divergence as a vulnerability indicator, decomposed into full-distribution, left-tail, and right-tail components following the convention introduced by Adrian, Boyarchenko, and Giannone (2019). Section 5 describes distributional skewness as a complementary asymmetry diagnostic. Section 6 details the computational configuration: the choice of horizons, confidence levels, window sizes, and retraining cadence, each grounded in a combination of regulatory guidelines and the empirical properties of the data. Section 7 outlines the backtesting framework that validates these risk measures against realised outcomes.

2 Value-at-Risk

Value-at-Risk is, at its core, a quantile. It answers a precise question: what is the threshold loss (or gain) that the portfolio will not exceed with probability $1 - \alpha$ over a holding period of h days? If $F_{t+h}(x)$ denotes the cumulative distribution function of the h -step-ahead return conditional on information available at date t , then the left-tail Value-at-Risk at confidence level $1 - \alpha$ is

$$\text{VaR}_{t+h}^{\text{Left}}(\alpha) = F_{t+h}^{-1}(\alpha), \quad (1)$$

and the right-tail counterpart, which we label *Growth-at-Risk* following the terminology of the framework, is

$$\text{VaR}_{t+h}^{\text{Right}}(1 - \alpha) = F_{t+h}^{-1}(1 - \alpha). \quad (2)$$

The sign convention is economically intuitive: $\text{VaR}^{\text{Left}} < 0$ represents a loss threshold,

and $\text{VaR}^{\text{Right}} > 0$ represents a gain threshold. The left-tail measure is the Oil-at-Risk in the vocabulary of this thesis; the right-tail measure is the Growth-at-Risk. Computing both is essential because the conditional distribution fitted by the Johnson SU is generically asymmetric. When the geopolitical risk index is elevated and the dollar is weakening, the left and right tails need not move in parallel; reporting only the left tail would miss the information contained in the right tail's response to the same conditioning variables.

Conditional Value-at-Risk from the Johnson SU

The conditional VaR follows directly from the fitted density. At each forecast origin t , the distribution fitting step has produced an estimated parameter vector $\hat{\Theta}_t = \{\hat{\gamma}_t, \hat{\delta}_t, \hat{\xi}_t, \hat{\lambda}_t\}$ for the Johnson SU distribution. The conditional Value-at-Risk is obtained by evaluating the quantile function at the desired tail probability:

$$\text{VaR}_{t+h}^{\text{Cond, Left}}(\alpha) = \hat{\xi}_t + \hat{\lambda}_t \sinh\left(\frac{\Phi^{-1}(\alpha) - \hat{\gamma}_t}{\hat{\delta}_t}\right), \quad (3)$$

where $\Phi^{-1}(\cdot)$ is the standard normal quantile function. This expression evaluates in microseconds and involves no numerical root-finding, inheriting the computational advantage of the SU quantile function discussed in the distribution fitting chapter.

A critical observation is that no time-scaling is needed for the conditional VaR. The quantile regression that produced the inputs to the distribution fitting step was estimated with the target variable y_{t+h} , the return over the next h trading days. The fitted Johnson SU distribution is therefore already a distribution of h -step-ahead returns. Its α -quantile is directly the h -day VaR, without any square-root-of-time adjustment. This stands in contrast to the unconditional approach described next, where the distribution is fitted to daily returns and must be scaled to the desired horizon.

Unconditional Value-at-Risk: Historical Simulation

The unconditional VaR serves as the benchmark against which the conditional measure is assessed. It is computed by Historical Simulation, the most widely used non-parametric VaR methodology in practice. The procedure is straightforward: at each evaluation date t , collect the most recent W daily returns $\{r_{t-W+1}, \dots, r_t\}$, compute the empirical α -quantile of this sample, and scale the result to the h -day horizon using the square-root-of-time rule:

$$\text{VaR}_{t+h}^{\text{Hist, Left}}(\alpha) = \hat{q}_\alpha(\{r_{t-W+1}, \dots, r_t\}) \times \sqrt{h}, \quad (4)$$

where $\hat{q}_\alpha$ denotes the empirical α -quantile. The square-root-of-time scaling is the standard convention in regulatory risk management, codified in Basel III §OIS20. It is exact when daily returns are independent and identically distributed with finite variance, because the variance of the sum of h i.i.d. draws is h times the variance of a single draw, and the

standard deviation therefore scales as $\sqrt{h}$. In practice, daily oil returns exhibit moderate serial dependence and time-varying volatility, so the scaling is an approximation, but it is the one mandated by the regulatory framework and used universally in the industry.

The Historical Simulation approach carries two well-known limitations that motivate the conditional alternative. First, it is entirely backward-looking: the empirical quantile reflects the distribution of returns over the estimation window, not the distribution conditional on today's information. A window that spans both a geopolitical crisis and a period of calm will produce a VaR that averages across the two regimes, understating risk during escalation and overstating it during tranquillity. Second, all observations within the window receive equal weight. A five-year-old return has the same influence on the quantile as yesterday's return, even though the volatility regime may have changed entirely in between. The conditional VaR from the Johnson SU suffers from neither limitation, because the quantile regression coefficients absorb the current state of the conditioning variables, and the distribution fitting produces a density that is specific to the forecast origin. The price of this conditioning is model risk: if the quantile regression is misspecified or the Johnson SU is an inadequate distributional family, the conditional VaR will be biased in ways that the model-free historical quantile is not. Reporting both measures side by side is therefore not redundant; it is a diagnostic strategy. When the two VaR series track each other closely, the conditioning variables are adding little information beyond what the recent history already contains. When they diverge, the conditional model is detecting a regime shift that the backward-looking historical measure has not yet absorbed into its window.

The window size W and the confidence level $1 - \alpha$ are set following Basel III guidelines, as discussed in the configuration section below.

3 Expected Shortfall

Value-at-Risk answers the question of how bad a loss can be at a given confidence level, but it is silent on what happens beyond that threshold. This is a consequential omission. Two distributions can share the same 97.5th percentile while having radically different tail behaviour beyond it: one may decay rapidly to zero, implying that exceedances beyond the VaR are rare and mild, while the other may exhibit a heavy tail extending far into the loss region, implying that when the VaR is breached, the resulting loss is likely to be catastrophic. In crude oil markets, this distinction is not hypothetical. The distribution of returns during the COVID-19 demand collapse of April 2020 had a left tail that extended to losses of 30 percent or more in a single day, while the distribution during the preceding months of gradual price decline had a much thinner left tail despite a similar VaR at the 2.5th percentile.

Expected Shortfall, also known as Conditional Value-at-Risk, addresses this limitation

by measuring the average loss in the tail beyond the VaR threshold. Formally, for the left tail,

$$\text{ES}_{t+h}^{\text{Left}}(\alpha) = \mathbb{E}[r_{t+h} \mid r_{t+h} \leq \text{VaR}_{t+h}^{\text{Left}}(\alpha)], \quad (5)$$

and for the right tail,

$$\text{ES}_{t+h}^{\text{Right}}(1 - \alpha) = \mathbb{E}[r_{t+h} \mid r_{t+h} \geq \text{VaR}_{t+h}^{\text{Right}}(1 - \alpha)]. \quad (6)$$

The left-tail Expected Shortfall is always more negative than the left-tail VaR, and the right-tail Expected Shortfall is always more positive than the right-tail VaR, because the conditional expectation in the tail lies beyond the tail's boundary by construction. The difference $\text{ES}^{\text{Left}} - \text{VaR}^{\text{Left}}$ measures the expected severity of a VaR exceedance, a quantity of direct interest to a risk manager who has already accepted that the VaR threshold may be breached and wants to know how bad the outcome is likely to be when it is.

Expected Shortfall enjoys a property that Value-at-Risk does not: it is a coherent risk measure in the sense of Artzner, Delbaen, Eber, and Heath (1999). The critical property is sub-additivity: the Expected Shortfall of a combined portfolio is never larger than the sum of the Expected Shortfalls of its components. Value-at-Risk can violate this property for non-elliptical distributions, which means that it can, perversely, assign a higher risk to a diversified portfolio than to the sum of its undiversified components. This theoretical shortcoming was one of the principal motivations behind the Basel Committee's decision, in the Fundamental Review of the Trading Book, to replace VaR with Expected Shortfall as the basis for internal-model capital calculations under Basel III.

Conditional Expected Shortfall from the Johnson SU

When the conditional density is available in closed form, the Expected Shortfall integral can be evaluated directly. For the Johnson SU distribution, no closed-form expression for the conditional tail expectation exists, so we compute it by numerical integration. The procedure evaluates the density on a fine grid of 5,000 points spanning the support of the distribution from the 10^{-6} quantile to the $1 - 10^{-6}$ quantile, isolates the portion of the grid below the VaR threshold (for the left tail) or above it (for the right tail), and computes the weighted average:

$$\text{ES}_{t+h}^{\text{Cond, Left}}(\alpha) = \frac{\int_{-\infty}^{\text{VaR}_{t+h}^{\text{Left}}(\alpha)} x f(x \mid \hat{\Theta}_t) dx}{\int_{-\infty}^{\text{VaR}_{t+h}^{\text{Left}}(\alpha)} f(x \mid \hat{\Theta}_t) dx} \approx \frac{\sum_{x_i \leq \text{VaR}_{t+h}^{\text{Left}}} x_i f(x_i \mid \hat{\Theta}_t) \Delta x}{\sum_{x_i \leq \text{VaR}_{t+h}^{\text{Left}}} f(x_i \mid \hat{\Theta}_t) \Delta x}, \quad (7)$$

where Δx is the grid spacing. As with the conditional VaR, no time-scaling is applied be-

cause the Johnson SU distribution already represents the h -step-ahead conditional density. The Riemann sum approximation on a 5,000-point grid introduces negligible numerical error relative to the statistical uncertainty in the estimated density parameters.

Unconditional Expected Shortfall: Historical Simulation

The unconditional Expected Shortfall is computed from the same rolling window of daily returns used for the historical VaR. Having identified the empirical α -quantile as the VaR threshold, the procedure isolates all returns in the window that fall below (or above, for the right tail) that threshold and computes their arithmetic mean:

$$\text{ES}_{t+h}^{\text{Hist, Left}}(\alpha) = \frac{1}{|\mathcal{T}_\alpha|} \sum_{r_s \in \mathcal{T}_\alpha} r_s \times \sqrt{h}, \quad \mathcal{T}_\alpha = \{r_s : r_s \leq \hat{q}_\alpha, s \in [t - W + 1, t]\}, \quad (8)$$

where $\mathcal{T}_\alpha$ is the set of returns in the left tail beyond the VaR threshold, and the square-root-of-time scaling is applied to aggregate from daily to h -day frequency under the same assumptions that justify the scaling for VaR.

The same comparative logic applies here as for VaR: the conditional and unconditional Expected Shortfall series are reported in parallel, and their divergence at any given date signals that the conditioning variables are detecting a tail-severity shift that the backward-looking historical measure has not yet registered.

4 Kullback–Leibler Divergence as a Vulnerability Indicator

Value-at-Risk and Expected Shortfall are point statistics extracted from the conditional density at specific tail probabilities. They answer questions about the boundary and depth of the tails, but they do not directly measure how much the conditional density has departed from a reference state. A risk manager who observes that the conditional VaR has moved from -3.2% to -3.8% knows that the tail threshold has shifted, but she does not know whether this shift reflects a localised movement of the 2.5th percentile or a wholesale reshaping of the entire conditional distribution. The distinction matters: a density that has shifted uniformly to the left, preserving its shape, poses a qualitatively different risk from one whose left tail has fattened while the centre and right tail remain unchanged.

Kullback–Leibler divergence provides a global measure of distributional distance that captures precisely this distinction. Given a conditional forecast density $p(x)$ and a reference density $q(x)$, the KL divergence from q to p is

$$D_{\text{KL}}(p \parallel q) = \int_{-\infty}^{\infty} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx. \quad (9)$$

The integrand $p(x) \ln(p(x)/q(x))$ is non-negative everywhere and equals zero only where $p(x) = q(x)$. The divergence is therefore always non-negative, and it equals zero if and only if the two densities are identical. It is not a metric in the mathematical sense, because it is asymmetric: $D_{\text{KL}}(p \parallel q) \neq D_{\text{KL}}(q \parallel p)$ in general. The direction of the asymmetry is economically meaningful. In the convention adopted here, $D_{\text{KL}}(p \parallel q)$ heavily penalises the reference density q when it assigns zero or near-zero probability to events that the conditional density p considers plausible. In risk management terms, this is precisely the dangerous scenario: the reference model says a large loss is essentially impossible, while the conditional forecast says it has non-negligible probability. The asymmetry of the KL divergence ensures that such model failures are penalised severely.

The KL divergence can be interpreted through the lens of information theory as the expected excess surprise when the reference density q is used to describe a world that is actually governed by p . Each realisation drawn from p carries $-\ln q(x)$ units of surprise under the reference model, but only $-\ln p(x)$ units under the correct model. The divergence is the expected value of this difference, taken under the true distribution p :

$$D_{\text{KL}}(p \parallel q) = \mathbb{E}_p \left[\ln \frac{p(x)}{q(x)} \right] = \mathbb{E}_p[-\ln q(x)] - \mathbb{E}_p[-\ln p(x)]. \quad (10)$$

The first term is the cross-entropy of q under p ; the second is the entropy of p itself. The divergence measures the extra information, in nats, needed to describe the true state of the world when the wrong model is assumed.

Tail Decomposition

The full-distribution KL divergence summarises the overall departure of the conditional density from the reference, but it does not reveal where the departure is concentrated. A large divergence driven entirely by a fattening of the left tail has very different economic implications from one driven by a fattening of the right tail, or by a shift in the centre. Following the convention introduced by Adrian, Boyarchenko, and Giannone (2019) in the Growth-at-Risk framework, we decompose the divergence into three components by restricting the integration domain.

Let $q_\tau = Q_q^{-1}(\tau)$ denote the τ -quantile of the reference density. The left-tail entropy is

$$D_{\text{KL}}^{\text{Left}}(p \parallel q) = \int_{-\infty}^{q_{1-\tau}} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx, \quad (11)$$

the right-tail entropy is

$$D_{\text{KL}}^{\text{Right}}(p \parallel q) = \int_{q_\tau}^{\infty} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx, \quad (12)$$

and the full divergence satisfies $D_{\text{KL}}^{\text{Full}} = D_{\text{KL}}^{\text{Left}} + D_{\text{KL}}^{\text{Centre}} + D_{\text{KL}}^{\text{Right}}$, where the centre component captures the divergence in the region between the two tail cutoffs.

The parameter τ controls how much of the distribution is allocated to the tails. In the original Growth-at-Risk framework, Adrian et al. use $\tau = 0.50$, so that the “left tail” is the entire lower half of the distribution and the “right tail” is the entire upper half. This choice is natural for quarterly GDP growth, whose distribution is relatively compact. For high-frequency commodity returns, however, the median split is too coarse: the interesting vulnerability signals are concentrated in the extreme tails, and a decomposition that assigns 50 percent of the distribution to each “tail” dilutes these signals with the behaviour of the centre. We therefore set $\tau = 0.95$ as the default, so that the left-tail entropy integrates only over the region below the 5th percentile of the reference density, and the right-tail entropy integrates only over the region above the 95th percentile. This choice isolates the extreme tails where supply disruptions and demand collapses manifest most acutely. An analysis using the Adrian et al. median decomposition is readily available as a robustness check by setting $\tau = 0.50$.

Choice of Reference Density

The KL divergence requires a reference density $q(x)$. We compute the divergence against two alternative baselines, each answering a different economic question.

The first baseline is a Normal density calibrated to the historical mean and standard deviation of returns over the estimation window. The divergence $D_{\text{KL}}(p_{\text{cond}} \parallel q_{\text{Normal}})$ measures how much the conditional forecast departs from Gaussianity, capturing the combined effect of skewness, excess kurtosis, and tail mass that the Normal density cannot represent. When this divergence is low, the conditional distribution is approximately Gaussian and standard delta-Normal risk measures are adequate. When it is high, the conditional distribution exhibits features that Gaussian-based risk management would miss entirely. This measure functions as an early warning system: a rapid increase in the entropy against the Normal baseline signals that the market’s return distribution is moving away from the well-behaved regime in which traditional risk metrics are reliable.

The second baseline is the unconditional Johnson SU density fitted by Maximum Likelihood to the historical returns over the same window. This is the baseline used in the original Vulnerable Growth framework of Adrian, Boyarchenko, and Giannone (2019). The divergence $D_{\text{KL}}(p_{\text{cond}} \parallel q_{\text{uncond}})$ measures how much the conditioning information has shifted the forecast density relative to what the historical distribution alone would predict. When this divergence is low, the conditioning variables are not contributing meaningful information beyond the historical baseline. When it is high, the conditional model is

detecting a distributional shift that the unconditional fit, averaged over all regimes in the window, has not yet absorbed. This baseline is more demanding than the Normal, because the unconditional JSU already accommodates skewness and heavy tails; a high divergence against it signals a genuine regime shift in the conditional distribution, not merely a departure from Gaussianity.

Both divergences are computed by Riemann summation on a 5,000-point grid spanning the joint support of the two densities, using the same numerical integration approach described for Expected Shortfall. The integrand is masked where either density falls below 10^{-12} to prevent numerical overflow in the logarithm.

5 Distributional Skewness

The Kullback–Leibler divergence and its tail decomposition capture the overall and directional departure of the conditional density from a reference, but they do not directly report the asymmetry of the forecast distribution itself. Distributional skewness provides this complementary diagnostic. The Fisher skewness coefficient (the standardised third central moment) of the fitted Johnson SU distribution is computed analytically using the closed-form expressions derived by Johnson (1949):

$$\gamma_1 = -\sqrt{\omega} (\omega + 2) \frac{3 \sinh\left(\frac{\gamma}{\delta}\right) + \sinh\left(\frac{3\gamma}{\delta}\right)}{4 \left[\frac{\omega-1}{2} (\omega \cosh\left(\frac{2\gamma}{\delta}\right) + 1) \right]^{3/2}}, \quad (13)$$

where $\omega = \exp(1/\delta^2)$ and γ, δ are the shape parameters of the fitted Johnson SU. The skewness is negative when the left tail is heavier than the right, signalling an asymmetric accumulation of downside risk, and positive when the right tail dominates, signalling upside asymmetry.

The economic content of time-varying skewness in crude oil markets is substantial. During periods of geopolitical escalation, when the market prices in the possibility of a supply disruption but not a demand collapse, the right tail of the return distribution fattens relative to the left, and skewness becomes positive or less negative. During demand-side recessions, the opposite pattern emerges. Tracking the conditional skewness over time therefore reveals the direction in which the balance of risks is tilted at each forecast origin, a diagnostic that the symmetric KL divergence against a Normal baseline cannot provide.

We compute both the conditional skewness, from the Johnson SU parameters estimated by Minimum Distance Estimation at each forecast origin, and the unconditional skewness, from the Johnson SU fitted by Maximum Likelihood to the historical returns up to each date. The difference between the two tracks the incremental asymmetry contributed by the conditioning variables.

6 Computational Configuration

The risk metrics described above are computed over an out-of-sample evaluation that traverses every trading day in the sample, producing a time series of conditional and unconditional risk measures at each forecast origin. The configuration of this evaluation involves several choices, each of which is grounded in either regulatory guidelines, the empirical properties of the data, or the methodological requirements of the direct forecasting framework.

Forecast Horizon

The forecast horizon h is set to match the horizons used in the direct forecasting step. We focus on the short-term horizons, specifically $h = 2$ trading days, because these are the horizons at which the quantile regression model exhibits the strongest predictive power, as demonstrated by the pinball loss comparisons in the direct forecasting chapter. Longer horizons are available and can be analysed within the same framework, but the accuracy of the conditional density deteriorates as h increases, because the quantile regression coefficients become noisier and the Minimum Distance Estimation inherits that noise. Reporting risk metrics at horizons where the underlying quantile forecasts are unreliable would undermine the credibility of the conditional measures.

Confidence Level

The confidence level for Value-at-Risk and Expected Shortfall is set at 97.5 percent, following the Basel III Fundamental Review of the Trading Book. The earlier Basel I and Basel II.5 frameworks used a 99 percent confidence level, but the Basel III revision lowered it to 97.5 percent for a reason directly relevant to this thesis. Under heavy-tailed distributions, which crude oil returns notoriously exhibit, estimating the 99th percentile with precision requires extremely large samples. The 1st percentile of a distribution with excess kurtosis of 5 or 6 is located far into the tail, where the density is thin and small perturbations in the data can shift the quantile substantially. Moving to the 97.5th percentile provides a more robust estimate while still capturing the tail of the distribution.

There is a second, methodological reason for not estimating the VaR directly from the quantile regression at the 97.5th or 99th percentile. Quantile regression at extreme quantiles is imprecise by construction: with a sample of W observations, the effective sample size for estimating the α -th quantile is approximately αW for the left tail and $(1 - \alpha)W$ for the right tail. At $\alpha = 0.025$, this means roughly 25 effective observations per 1,000. At $\alpha = 0.01$, it drops to 10. Fitting a regression with meaningful standard errors on 10 effective observations is statistically dubious. The Oil-at-Risk framework circumvents this limitation by fitting the quantile regression at moderate quantile levels,

specifically $\tau \in \{0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99\}$, recovering a continuous density via the Johnson SU, and then reading the 2.5th percentile from the fitted density. The density interpolates between the estimated quantiles, extracting tail information from the shape of the distribution rather than from a single regression at an extreme quantile level. This two-step procedure, quantile regression followed by distributional fitting, is precisely what makes the framework capable of producing reliable risk measures at confidence levels that a direct quantile regression at $\tau = 0.025$ could not sustain.

Window Sizes

Two distinct window sizes govern the computation. The rolling window for the historical VaR and CVaR is set at $W = 1,008$ observations, corresponding to four trading years of 252 days each. This is the standard calibration window prescribed by Basel III for internal-model VaR calculations. Four years provides enough history to capture a range of volatility regimes while remaining short enough that the oldest observations in the window are still plausibly relevant to the current market environment.

The rolling window for the conditional density estimation, which feeds the KL divergence computation, uses the same window size as the direct forecasting evaluation. The conditional quantile regressions are fitted on the most recent W observations, and the Minimum Distance Estimation produces a density specific to each forecast origin. Using the same window for both the conditional and unconditional measures ensures that the comparison between them is not confounded by differences in sample coverage.

Expanding versus Rolling Windows

The out-of-sample evaluation can be conducted with either an expanding window, which grows from the beginning of the sample to each forecast origin, or a rolling window, which uses only the most recent W observations and discards older data. Both approaches are implemented in the codebase.

The expanding window is the canonical choice in the Growth-at-Risk literature: Adrian, Boyarchenko, and Giannone (2019) use expanding windows throughout their empirical exercise. Its advantage is that every available observation contributes to the estimate, maximising the information content and reducing the variance of the fitted parameters. Its disadvantage is that distant observations receive the same weight as recent ones, which can be problematic in markets with structural breaks or regime changes. A model estimated on an expanding window that includes the 2008 financial crisis and the 2014–2016 oil glut will carry the imprint of those regimes even when current conditions bear no resemblance to them.

The rolling window discards the oldest observations as new ones enter, so the model continually adapts to the most recent regime. This is advantageous when the data-

generating process changes over time, as is manifestly the case in crude oil markets. The cost is a reduction in effective sample size: with $W = 1,008$, the model sees only the last four years of data at any given origin. In practice, this is more than sufficient for estimating quantile regressions with a small number of regressors, and the gain in adaptability outweighs the loss in precision. Additionally, the rolling window reduces computation time substantially, because each iteration processes W observations rather than a growing fraction of the full sample.

Both window types are available for the entropy and skewness computations, and the results can be compared to assess the sensitivity of the vulnerability measures to the treatment of distant history.

Retraining Cadence

The quantile regression coefficients that underpin the conditional density are not re-estimated at every forecast origin. Instead, the coefficients are refitted once every 30 out-of-sample observations, approximately one trading month, and held fixed in between. On the intervening days, the cached regression coefficients are applied to the current day’s predictor values, so the conditional density still updates daily through the changing co-variates; only the mapping from covariates to quantiles is held constant between retraining dates.

This choice reflects a balance between statistical freshness and computational feasibility. Re-estimating the full set of quantile regressions at every origin, which requires solving a linear programme for each quantile level, is the theoretically purest approach but imposes a computational burden that scales linearly with the number of out-of-sample dates. The 30-day cadence reduces this burden by a factor of 30 while preserving the daily variation in the conditional density through the covariate channel. The calibration was chosen to be conservative: macroeconomic conditions that affect the oil market, such as geopolitical escalation or monetary policy shifts, evolve on timescales longer than a month, so the regression coefficients estimated at one retraining date are unlikely to have become materially stale by the next. A robustness check with daily retraining (`retrain_after = 1`) confirms that the risk metric series are nearly indistinguishable, validating the cadence choice.

KL Divergence Tail Mark

The tail decomposition of the Kullback–Leibler divergence uses $\tau = 0.95$ as the default mark, so that the left-tail entropy integrates below the 5th percentile and the right-tail entropy integrates above the 95th percentile of the reference density. As noted in Section 4, this differs from the $\tau = 0.50$ median split used by Adrian, Boyarchenko, and Giannone (2019). The choice is motivated by the high-frequency nature of the data: daily oil

returns exhibit substantially more intraday noise and less persistent tail behaviour than the quarterly GDP growth rates studied in the original Growth-at-Risk framework. Concentrating the tail entropy on the extreme 5 percent of each tail provides a sharper signal of genuine vulnerability, filtering out the fluctuations in the centre of the distribution that are economically uninteresting for risk management purposes. The median decomposition remains available as an alternative specification, and its results are qualitatively similar but noisier.

7 Backtesting Framework

Risk measures are only as credible as the evidence that they perform as advertised. A Value-at-Risk at the 97.5 percent confidence level is, implicitly, a promise: in no more than 2.5 percent of periods should the realised loss exceed the stated threshold. If exceedances occur materially more often, the model is underestimating risk and the capital reserves it implies are insufficient. If exceedances are materially rarer, the model is overly conservative and the capital it reserves is unnecessarily large, imposing an economic cost in the form of foregone investment.

Backtesting is the ex-post verification of this promise. It compares the sequence of predicted risk measures against the sequence of realised returns and asks whether the observed pattern of exceedances is consistent with the model's stated coverage. The framework implements three progressively more stringent tests for VaR and two specialised tests for Expected Shortfall.

Kupiec Unconditional Coverage Test

The Kupiec (1995) test is the simplest and most widely used VaR backtest. It counts the number of times the realised return exceeds the VaR threshold over the evaluation period and tests whether this count is consistent with the nominal exceedance probability $\alpha = 1 - \text{confidence}$. Under the null hypothesis that the model is correctly calibrated, the number of exceedances follows a binomial distribution with parameters n (the number of evaluation dates) and α . The test statistic is a likelihood ratio:

$$LR_{UC} = -2 \ln \left[\frac{\alpha^x (1 - \alpha)^{n-x}}{\hat{\pi}^x (1 - \hat{\pi})^{n-x}} \right], \quad \hat{\pi} = \frac{x}{n}, \quad (14)$$

where x is the observed number of exceedances and $\hat{\pi}$ is the empirical exceedance rate. Under H_0 , the statistic is asymptotically $\chi^2(1)$. Rejection indicates that the model's stated coverage does not match the realised frequency of exceedances, but the test is agnostic about whether the exceedances are clustered or dispersed.

Christoffersen Conditional Coverage Test

The Christoffersen (1998) test augments the Kupiec test by also examining whether exceedances are independent over time. A model that produces the correct overall exceedance rate but clusters its exceptions, for example generating several consecutive VaR breaches during a crisis followed by long stretches without any, is poorly calibrated even if the total count is right. Such clustering violates the assumption of serial independence that underpins the regulatory use of VaR.

The Christoffersen test decomposes the likelihood ratio into an unconditional coverage component (equivalent to the Kupiec test) and an independence component that tests whether the probability of an exceedance on day t depends on whether an exceedance occurred on day $t - 1$. The combined test statistic is $LR_{CC} = LR_{UC} + LR_{Ind}$, distributed as $\chi^2(2)$ under the null of correct unconditional coverage and independence.

Engle–Manganelli Dynamic Quantile Test

The Dynamic Quantile test of Engle and Manganelli (2004) is the most powerful of the three VaR backtests implemented. It regresses the hit sequence, the binary indicator of VaR exceedance centred at zero, on a set of lagged hits and the contemporaneous VaR forecast. Under the null of correct model calibration, the hit sequence should be unpredictable: neither its own past values nor the level of the VaR forecast should help predict whether an exceedance will occur. A significant F-statistic indicates that the model's exceptions are systematically related to observable variables, a sign of model misspecification.

Basel Traffic Light System

The Basel Traffic Light system maps the number of VaR exceedances observed over the most recent 250 trading days into a Green, Yellow, or Red zone using the binomial distribution under the null of correct calibration. The zone boundaries are calibrated so that a correctly specified model falls in the Green zone with high probability, in the Yellow zone with moderate probability indicating potential concern, and in the Red zone with very low probability indicating a serious model failure. Models in the Red zone trigger mandatory capital multiplier add-ons under the regulatory framework.

Expected Shortfall Backtesting

Backtesting Expected Shortfall is fundamentally more challenging than backtesting VaR because ES is not elicitable in the classical sense: there is no loss function whose expectation is uniquely minimised at the true ES. This means that the straightforward exceedance-counting approach that works for VaR has no direct analogue for ES.

Acerbi and Székely (2014, 2017) resolved this challenge by exploiting the structure of ES as a tail average rather than a quantile breach count. Their tests use the joint elicibility of VaR and ES, constructing test statistics based on the severity of VaR exceedances rather than merely their frequency. If the ES is correctly specified, the average loss conditional on a VaR breach should equal the stated ES. Their non-parametric, model-independent tests do not require distributional assumptions about the return process, making them robust to the heavy tails and time-varying volatility that characterise oil markets.

The Costanzino and Curran (2015, 2018) ES Traffic Light test extends the Basel zone framework from VaR to ES, using a generalised coverage statistic that incorporates breach severity. The zone boundaries are calibrated under the FRTB parameters of $\alpha = 2.5\%$ and $N = 250$ observations, providing a direct regulatory analogue to the VaR traffic light.

Together, these backtesting procedures provide a comprehensive evaluation of both the threshold accuracy (VaR) and the tail-severity accuracy (ES) of the Oil-at-Risk framework’s risk measures.

References

- Acerbi, C. and Székely, B. (2014). Backtesting expected shortfall. *Risk Magazine*.
- Adrian, T., Boyarchenko, N. and Giannone, D. (2019). Vulnerable Growth. *American Economic Review*, 109(4), 1263–1289.
- Artzner, P., Delbaen, F., Eber, J.-M. and Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228.
- Basel Committee on Banking Supervision (2019). *Minimum capital requirements for market risk*. Basel III, §OIS20.
- Christoffersen, P. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Costanzino, N. and Curran, M. (2018). A simple traffic light approach to backtesting expected shortfall. *Risks*, 6(1), 2.
- Engle, R. F. and Manganelli, S. (2004). CAViaR: Conditional Autoregressive Value at Risk by Regression Quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- Johnson, N. L. (1949). Systems of frequency curves generated by methods of translation. *Biometrika*, 36(1/2), 149–176.
- Kupiec, P. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.